APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9

First Stage Landing Prediction

Predicting rocket landing outcomes using Machine Learning — SVM, Decision Trees & KNN

3 ML Models	**94.44%** Best Accuracy	**2** Data Sources	**5** Analysis Stages

Md. Masum Khan | March 2026

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EXECUTIVE SUMMARY

Best Model

Decision Tree

94.44% accuracy

Runner-Up

SVM & KNN

83.33% accuracy

Top Launch Site

KSC LC-39A

41.7% of successes

Data Collection

SpaceX API + Wikipedia web scraping → Integrated dataset of Falcon 9 launches.

Data Processing

Cleaned, wrangled and stored in Db2. SQL queries for exploratory insights.

Feature Engineering

Created new features, standardized data, and prepared for ML pipeline.

Visual Analytics

Folium maps for geospatial analysis; Plotly Dash for interactive dashboard.

ML Modeling

SVM, Decision Trees, KNN with GridSearchCV tuning. Decision Tree = best performer.

INTRODUCTION

Project Background & Context

SpaceX significantly reduces launch costs through Falcon 9 first-stage reuse. Accurately predicting landing success enables cost estimation and empowers competitors to strategically bid against SpaceX for contracts.

Research Questions

- **Q1** What factors influence Falcon 9 first stage landing success?
- **Q2** How can ML models accurately predict landing outcomes?
- **Q3** Which model performs best for binary classification?

METHODOLOGY

End-to-End Data Science Pipeline

1

Data Collection

SpaceX REST API + Wikipedia web scraping (BeautifulSoup)

2

Data Wrangling

Missing value imputation, normalization, feature engineering

3

EDA & Visualization

Matplotlib, Seaborn, SQL queries on Db2 database

4

Interactive Analytics

Folium geospatial maps + Plotly Dash dashboard

5

Predictive Modeling

SVM, Decision Tree, KNN with GridSearchCV tuning

DATA COLLECTION

SpaceX REST API

Primary Source

Step 1

`requests.get()`

Initiate API call to SpaceX v4/launches endpoint

Step 2

`response.json()`

Parse JSON → extract date, site, payload, outcome

Step 3

`pandas.DataFrame()`

Store as structured DataFrame

Step 4

`data.to_csv()`

Persist locally for offline processing

Wikipedia Scraping

Secondary Source

Step 1

`requests.get()`

Fetch Wikipedia Falcon 9 launches page HTML

Step 2

`BeautifulSoup()`

Parse HTML and locate launch records table

Step 3

`pandas.read_html()`

Convert table to DataFrame

Step 4

`Data Cleaning`

Standardize columns, drop nulls, merge with API data

DATA WRANGLING

01

Data Cleaning

- Identify and handle missing values (fillna / dropna)
- Remove duplicate records
- Standardize text fields (lowercase, strip whitespace)

02

Data Transformation

- Convert date strings → datetime format
- Cast numerical columns to correct types
- Normalize/scale features with MinMaxScaler or StandardScaler

03

Feature Engineering

- Extract year, month from launch dates
- Create binary target variable (landing success)
- Encode categorical variables for ML pipeline

04

Integration & Validation

- Merge SpaceX API + Wikipedia DataFrames
- Align column names and formats across sources
- Final consistency checks before analysis

EXPLORATORY DATA ANALYSIS

Visualization Techniques Used

Histogram

Purpose:

Distribution of payload mass, flight number, and launch success rates

Why:

Identifies data spread, outliers, and skewness

Bar Chart

Purpose:

Launch outcomes compared across sites and rocket types

Why:

Highlights frequency patterns in categorical data

Line Chart

Purpose:

Year-over-year trend in Falcon 9 launch success rates

Why:

Reveals temporal performance improvements

Scatter Plot

Purpose:

Payload mass vs. landing success relationship

Why:

Uncovers correlations between continuous variables

Heatmap

Purpose:

Correlation matrix across all numerical features

Why:

Identifies multicollinearity, guides feature selection

Box Plot

Purpose:

Spread and quartile distribution of numerical data

Why:

Highlights outliers and cross-category comparisons

EDA WITH SQL

Structured Query Analysis

Aggregate	Total launches, success/failure counts, success rates by site & rocket type
JOIN Queries	Linked launch records with rocket details for comprehensive cross-table analysis
Filtering	Extracted subsets by outcome, date range, and rocket configuration criteria
Sorting	Ordered launches by date and success rate to identify trends and outliers
Subqueries	Derived metrics such as average payload mass per launch site using nested queries

INTERACTIVE ANALYTICS

Folium Maps · Plotly Dash

🗺 Folium — Geospatial Maps

Markers	Pinpointed all launch sites on interactive world map
Circles	Illustrated proximity zones & safety perimeters
PolyLines	Measured distance to coast (~0.51 km for CCAFS SLC-40)
Cluster Map	26 launches at CCAFS LC-40: 19 failures, 7 successes

📊 Plotly Dash — Dashboard

Site Dropdown	Filter analysis by specific launch site
Payload Slider	Dynamically explore payload mass ranges
Pie Chart	Success vs. failure distribution at selected site
Scatter Plot	Payload mass vs. launch outcome correlation

EDA KEY FINDINGS

Insights from Charts and Visual Analytics

Flight Number vs. Launch Site

Consistent launch activity across all flight numbers at CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E. No single site dominates; mixed outcomes throughout.

Payload vs. Launch Site

KSC LC-39A handles the heaviest payloads (>15,000 kg) — preferred site for high-capacity missions. CCAFS SLC-40 typically below 10,000 kg.

Orbit Success Rates

VLEO, ES-L1, GEO, HEO, and SSO orbits achieved perfect 100% landing success. GTO orbit shows significantly lower success, indicating greater mission complexity.

Yearly Success Trend

Launch success rose sharply from 2013 onwards, exceeding 80% by 2020. Despite a dip in 2018, the overall trend confirms iterative improvement in landing reliability.

Payload vs. Orbit Type

Landings are most frequent for payloads below 6,000 kg. Above 10,000 kg shows mixed results, indicating payload mass increases mission risk.

PREDICTIVE ANALYSIS

Classification Modeling Pipeline

1. **Data Preprocessing**

   StandardScaler normalization + 80/20 train-test split for robust validation

2. **Model Selection**

   SVM · Decision Tree · K-Nearest Neighbors (KNN) · Logistic Regression

3. **Hyperparameter Tuning**

   GridSearchCV: C (SVM), max_depth (DT), n_neighbors (KNN)

4. **Model Evaluation**

   Accuracy, Precision, Recall, F1-Score, Confusion Matrix analysis

5. **Best Model**

   Decision Tree — 94.44% test accuracy, zero false negatives

MODEL PERFORMANCE RESULTS

🏆 BEST MODEL

Decision Tree

94.44%

Test Accuracy

Best performer. Zero false negatives — all successful landings correctly identified.

SVM

83.33%

Test Accuracy

Strong generalization. Good precision on binary classification task.

K-Nearest Neighbors

83.33%

Test Accuracy

Comparable to SVM. Performance depends on optimal k value via GridSearch.

Logistic Regression

83.33%

Test Accuracy

Solid baseline. Interpretable coefficients for feature importance analysis.

Confusion Matrix Insight (Decision Tree): 94.44% accuracy · 0 False Negatives · 1 False Positive — highly reliable for aerospace cost estimation and safety planning.

CONCLUSIONS

Key Findings & Recommendations

01

Launch Site Performance

CCAFS LC-40 leads in success rate (43.7%). KSC LC-39A accounts for 41.7% of all successful launches — both sites demonstrate optimal operational conditions.

02

Booster Version Reliability

"FT" booster version shows consistently high success across varied payload masses, demonstrating robust engineering — recommended for future high-priority missions.

03

Payload Mass Impact

No clear inverse correlation between payload mass and success rate found. Launch site and booster version are stronger predictors of landing outcome than payload alone.

04

Interactive Visualization Value

Folium and Plotly Dash delivered deep operational and geospatial insights, empowering stakeholders to make data-driven decisions at a glance.

Thank You

SpaceX Falcon 9 Landing Prediction — Data Science Capstone

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IBM Data Science Professional Certificate — Capstone Project

Decision Tree: 94.44%

SVM & KNN: 83.33%

KSC LC-39A: Best Site

Data: SpaceX API + Wikipedia